



VAULTX CYBER TECH

BEYOND TODAY · BEFORE TOMORROW

SUMMER INTERNSHIP PROGRAM 2026

WEEK

02

PROMPT ENGINEERING & STRUCTURED OUTPUT

PROMPT PATTERNS / SYSTEM PROMPTS / JSON / SCHEMA / FEW-SHOT / PROMPT EVALUATION

6

TASKS

6

DELIVERABLES

BEGINNER

LEVEL

ONSITE / LAB

MODE

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WEEK FOCUS

Objective	Get reliable, machine-parseable output from an LLM so it can drive real automations — strict JSON, schemas and evaluation. This is the backbone of every automation you build later.
Skills Targeted	Prompt patterns System prompts JSON / schema output Few-shot Prompt evaluation
Tools & Stack	Anthropic / OpenAI SDK Pydantic JSON schema your Week-01 API wrapper

TASK LIST

01 Master the core prompt patterns

- Build a reusable template library covering zero-shot, few-shot, role/system, chain-of-thought and constrained output.

02 Force structured output

- Make the model return strict JSON that matches a Pydantic schema.
- Validate the response and auto-retry when the JSON is invalid.

03 Build a classification tool

- Given an input (e.g. a support message), output {category, priority, sentiment, needs_human} as validated JSON.
- Test it on 20 realistic samples.

04 Build an extraction tool

- Extract structured fields from messy text (invoices, emails) into JSON.
- Handle missing fields gracefully instead of crashing.

05 Evaluate your prompts

- Build a tiny eval set (10–20 cases) with expected outputs.
- Measure accuracy, iterate the prompt, and record the improvement.

06 Package as a module

- Wrap these into reusable, logged functions with error handling that later automations can import.



DELIVERABLES

#	DELIVERABLE	FORMAT
1	Prompt template library	PY / MD
2	JSON-schema tool with validation + retry	PY
3	Classifier + evaluation results	PY + PDF
4	Field-extraction tool	PY
5	Eval harness + accuracy report	PY + PDF
6	Reusable structured-output module	PY

SUBMISSION PROTOCOL

Channel	Upload to the VaultX LMS under Internship > AI Track > Week 02.
File Naming	VX_AI_W02_FirstName_LastName.pdf plus one ZIP with all supporting files, exports and screenshots.
Deadline	Sunday, 11:59 PM (PKT) of the assigned week.
Evidence	Every screenshot must show your machine clock and the relevant page/scope.
Walkthrough	Record a short screen video (3–5 min) explaining what you did, what you found, and how you stayed safe.